

Abdul Khurram

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EDUCATION & AWARDS

University of British Columbia

Vancouver, BC, Canada

Bachelor of Science in Computer Science (Expected Graduation: May 2027)

Euclid Mathematics Contest, University of Waterloo, Top 5% Internationally

2021

WORK EXPERIENCE

Shopify

Sept. 2026 – Present

Software Engineer Intern

Toronto, ON, Canada

- Built a **Codex integration adopted across several teams**, packaging 9 reusable skills and session hooks with sandbox-validated installation and upgrade flows.
- Designed an **MCP bridge** to an internal agent, using API experiments to replace full-session scans with direct lookups and document identity and loop-control requirements.

RBC Royal Bank | *Python, Apache Spark, Airflow, SQL, Linux*

Jan. – Aug. 2026

Software Developer Co-op

Vancouver, BC, Canada

- Built Python/Airflow pipelines processing **50M+ daily transactions** from petabyte-scale Delta Lake, delivering analysis-ready data for downstream ML and analytics workloads.
- Optimized **Spark** jobs to cut **test runtime 40%**; rolled back the change after production instability.
- Worked on **multi-region pipelines** with observability, fault tolerance, idempotent re-runs, and automated failover for critical banking workloads.

Contello.ai | *TypeScript/Node.js, AWS, OpenAI, Playwright*

Jan. – Dec. 2025

AI Systems & Full-Stack Engineering Intern

Toronto, ON, Canada

- Built TypeScript/Node.js services and async AWS pipelines (SQS, S3) processing **thousands of content jobs/day**, owning features from API design through production monitoring.
- Integrated the **OpenAI Responses API** and an NLP/embedding context pipeline for brand-aware generation, with validation, responsible-AI guardrails, and cost/latency tracking.
- Raised scheduled-post reliability to **99.9%** through retry/backoff, CloudWatch alerting, Playwright E2E/integration tests, and CI/CD release gates.

RESEARCH EXPERIENCE

UBC Emerging Media Lab | *Unreal Engine, C++, Python*

Sept. – Dec. 2024

Software Developer & Team Lead

Vancouver, BC, Canada

- Led an interdisciplinary CS × Nursing team from requirements through delivery of an **AI-driven VR clinical simulation**.
- Built a **multithreaded C++/Python supervisor** for concurrent AI actors and protocol enforcement; optimized inference to cut **latency 40% and costs 90%**.

PROJECTS

Backer | *TypeScript, Python, AWS DynamoDB, XGBoost*

Hack The Coast 2026

- Architected a DynamoDB data layer (**10+ tables**) with typed repositories, batch operations, and transactional writes for high-throughput feed and messaging workflows.
- Engineered six behavioral and similarity features and trained an **XGBoost ranker** to personalize startup discovery from interaction signals.

Workbase | *Next.js/TypeScript, PostgreSQL, AWS Bedrock*

Personal Project

- **Solo-built** a full-stack platform that ingests GitHub repositories via OAuth, clusters evidence into work themes, and generates artifacts exclusively from **human-approved claims**.
- Engineered a **resource-bounded AWS Bedrock agent** capped at 8 iterations, 16 tool calls, and 120K tokens/run, with automated **credential redaction** before model calls.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, TypeScript/JavaScript, SQL, Bash

AI/ML: XGBoost, OpenAI API, AWS Bedrock, NLP/embeddings, model evaluation

Infrastructure & Data: Apache Spark, Airflow, Delta Lake; AWS (DynamoDB, S3, SQS, CloudWatch); PostgreSQL, Docker, Git, Linux, CI/CD